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Dataannotation Handbook

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Introduction

The purpose of this handbook is to guide the annotation of skin lesion images with respect to image quality. This is a collaboration between [omitted for blind review] and [omitted for blind review].

Annotation Process

The data to be annotated are images of skin lesions, and the evaluation will be performed manually, following the criteria established in the next section. It is important that each annotator evaluate all images individually to avoid bias and influence from other opinions.

Annotation Guide

To assess image quality, the following criteria should be considered:

- Blur / Focus
- Brightness
- Distance
- Framing
- External Interference
- General Perception

Each criterion must receive a score from 1 to 5 (the scores must be integers), where 1 is the worst and 5 is the best for that criterion. To standardize interpretation, the following qualitative scale applies to each criterion:

1️⃣ **Very Poor / Inadequate Image**: terrible, unusable.

2️⃣ **Poor / Below Average / Insufficient**: weak, compromised, of little use.

3️⃣ **Fair / Acceptable / Average**: reasonable, usable with limitations.

4️⃣ **Good / Adequate / Satisfactory**: sharp, reliable, well captured.

5️⃣ **Excellent / Great / Ideal**: perfect, clear, high-quality.

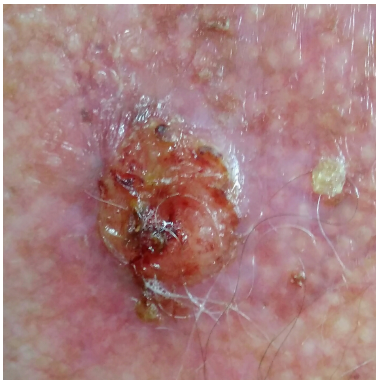
Explanation of Criteria

1 - Blur / Focus

This criterion evaluates the sharpness and focus of the image. Below are examples to guide this evaluation:

1️⃣ **Very Poor / Inadequate:** blurred, illegible, no definition.
2️⃣ **Poor / Below Average / Insufficient:** compromised focus, unclear edges.
3️⃣ **Fair / Acceptable / Average:** slight blur, but the lesion is still recognizable.
4️⃣ **Good / Adequate / Satisfactory:** sharp, main details preserved.
5️⃣ **Excellent / Great / Ideal:** perfect focus, microtextures visible.

Score: 1



Score: 5

Score: 4

Score: 3

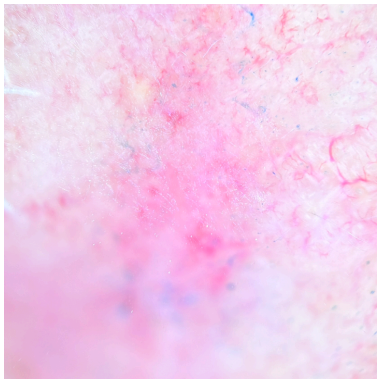
Score: 2

Example 1: blur / focus. Examples of scores for this criterion.

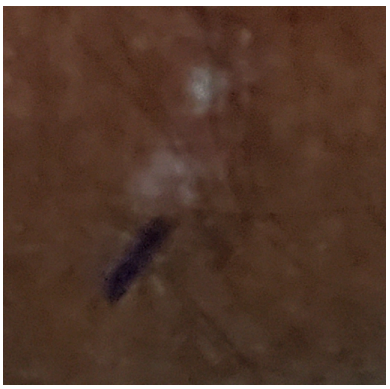
2 - Brightness

This criterion evaluates the lighting of the image. Images that are too dark or too bright, making it difficult to perceive the lesion, should be penalized.

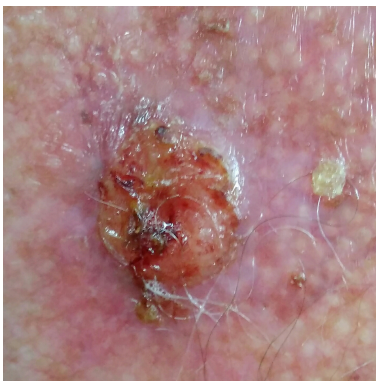
1️⃣ **Very Poor / Inadequate:** overexposed or underexposed, lesion not visible.
2️⃣ **Poor / Below Average / Insufficient:** irregular lighting, strong shadows or glare.
3️⃣ **Fair / Acceptable / Average:** acceptable lighting with minor flaws.
4️⃣ **Good / Adequate / Satisfactory:** balanced lighting, natural color.
5️⃣ **Excellent / Great / Ideal:** uniform light, true colors, no artifacts.



Score: 1

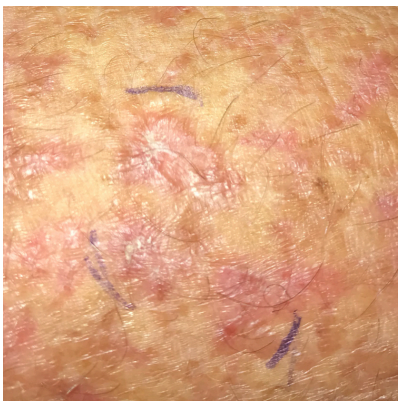


Score: 2



Score: 5

Score: 3



Score: 4

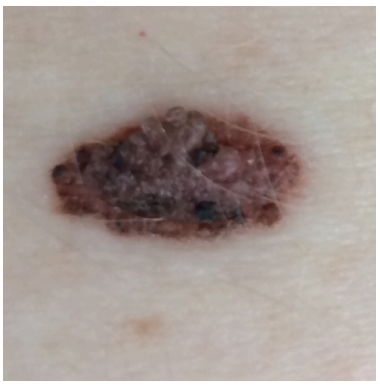
Example 2: brightness. Examples of scores for this criterion. The top-left image is too bright, and the top-middle image is too dark — both are considered poor.

3 - Distance

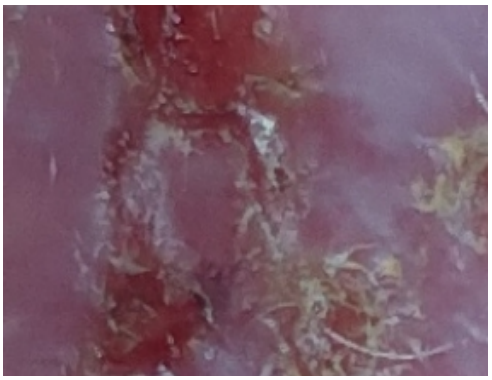
This criterion evaluates the distance between the camera and the lesion. Images that are too far and do not focus on the lesion should be penalized, while closer images that clearly show the lesion area should receive higher scores.

1️⃣ **Very Poor / Inadequate:** lesion is tiny or missing.
2️⃣ **Poor / Below Average / Insufficient:** too distant or excessively zoomed.
3️⃣ **Fair / Acceptable / Average:** reasonable proportion with some detail.
4️⃣ **Good / Adequate / Satisfactory:** correct framing, lesion clearly visible.
5️⃣ **Excellent / Great / Ideal:** perfect balance between detail and context.

Score: 1



Score: 5



Score: 2

Score: 1

Score: 4

Score: 3

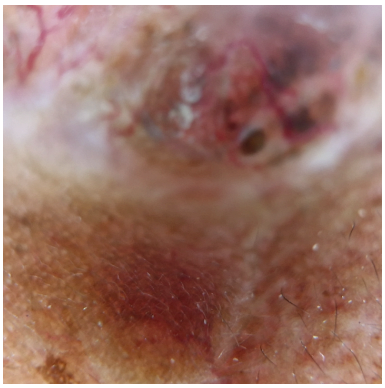
Example 3: Distance. The top-left image is too distant and does not focus on the lesion, while the top-right image is considered excellent because the lesion region is clearly identifiable.The middle-right image is too distant and the middle-left one is too close, possibly cutting part of the lesion.

4 - Framing

This criterion evaluates the positioning of the lesion within the image. Images that do not focus on the area of interest or cut part of the lesion should be considered poor. Images that show the lesion completely and clearly should receive higher scores.

1️⃣ **Very Poor / Inadequate:** lesion cropped, obstructed, or off-center.

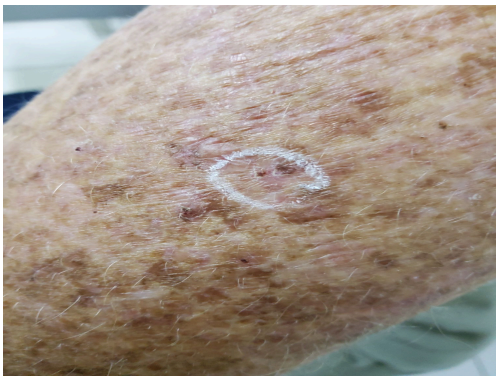
2️⃣ **Poor / Below Average / Insufficient:** visible artifacts, poor centering.
3️⃣ **Fair / Acceptable / Average:** minor misalignment, but usable.
4️⃣ **Good / Adequate / Satisfactory:** centered lesion, with artifacts or background that doesn't disturb the lesion visualization.
5️⃣ **Excellent / Great / Ideal:** precise framing, no distractions.



Score: 1

Score: 1

Score: 5



Score: 4

Score: 2

Score: 3

Example 4: framing. The top-left image shows the lesion poorly positioned, possibly cutting part of it. The top-right image does not focus precisely on the lesion region. The middle-left image is considered excellent for this criterion, while the middle-right image is an example of one that is good but still has some minor external interference (the background).

5 - External Interference

This criterion evaluates interference from external elements, such as other body parts, objects, or backgrounds with colors that contrast with the person's skin tone. Images containing distracting elements (e.g., rulers, pronounced pen marks, body hair, etc.) that obstruct lesion visibility should be penalized. Additionally, consider whether the background has a non-neutral color that interferes with the assessment. Anything not related to the lesion itself can negatively impact the classification.

1️⃣ **Very Poor / Inadequate:** lesion completely obstructed or overshadowed by background.
2️⃣ **Poor / Below Average / Insufficient:** object or background partially interferes.
3️⃣ **Fair / Acceptable / Average:** background or object present, but lesion clearly visible.
4️⃣ **Good / Adequate / Satisfactory:** no obstruction, clean/neutral background.
5️⃣ **Excellent / Great / Ideal:** lesion fully clear, no objects or distracting background.

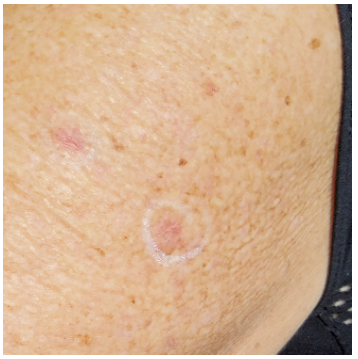
Score: 1

Score: 1

Score: 2

Score: 3

Score: 5



Score: 4

Example 5: external interference. The images at the top contain elements that hinder lesion visibility, such as a ruler or background, and are therefore considered to be of low quality. The bottom-left image can be considered fair because of the blue mark, which seems to have little influence, but mainly due to the hair interfering with the lesion's visibility. The bottom-right image is an example of a good image that still has minor interference (the background). The image outlined in green is an

excellent example — the hair has been intentionally moved aside to make the lesion clearly visible. It is important to note that, in this image, because the lesion is located in an area where hair is naturally present, it is not possible to eliminate this element completely; moving it aside to expose the lesion fully represents the best achievable condition and aligns with the “excellent” criteria.

To distinguish between scores 4 and 5, consider the following:

Score 5: no external elements present, or external elements are present but have been intentionally moved aside or arranged so that they do not interfere with lesion visibility.

Score 4: minor external elements are visible but they do not interfere with the visibility. Nothing obstructs the lesion, but the image is not completely ideal. These external objects or background elements could have been avoided during image capture, but since they appear in the frame, the image is assigned a score of 4 instead of 5.

The difference between this criterion and the framing criterion is as follows:

Framing: This criterion evaluates how well the lesion is centered and contained within the image. In some cases, part of the lesion may be cut off or missing.

External Interference: This refers to the presence of elements that are not part of the lesion itself. Examples include:

1. Ink markings (these are common, but sometimes they are too strong or overlap the lesion);
2. A ruler or pen pointing toward the lesion;
3. Backgrounds that are not skin.

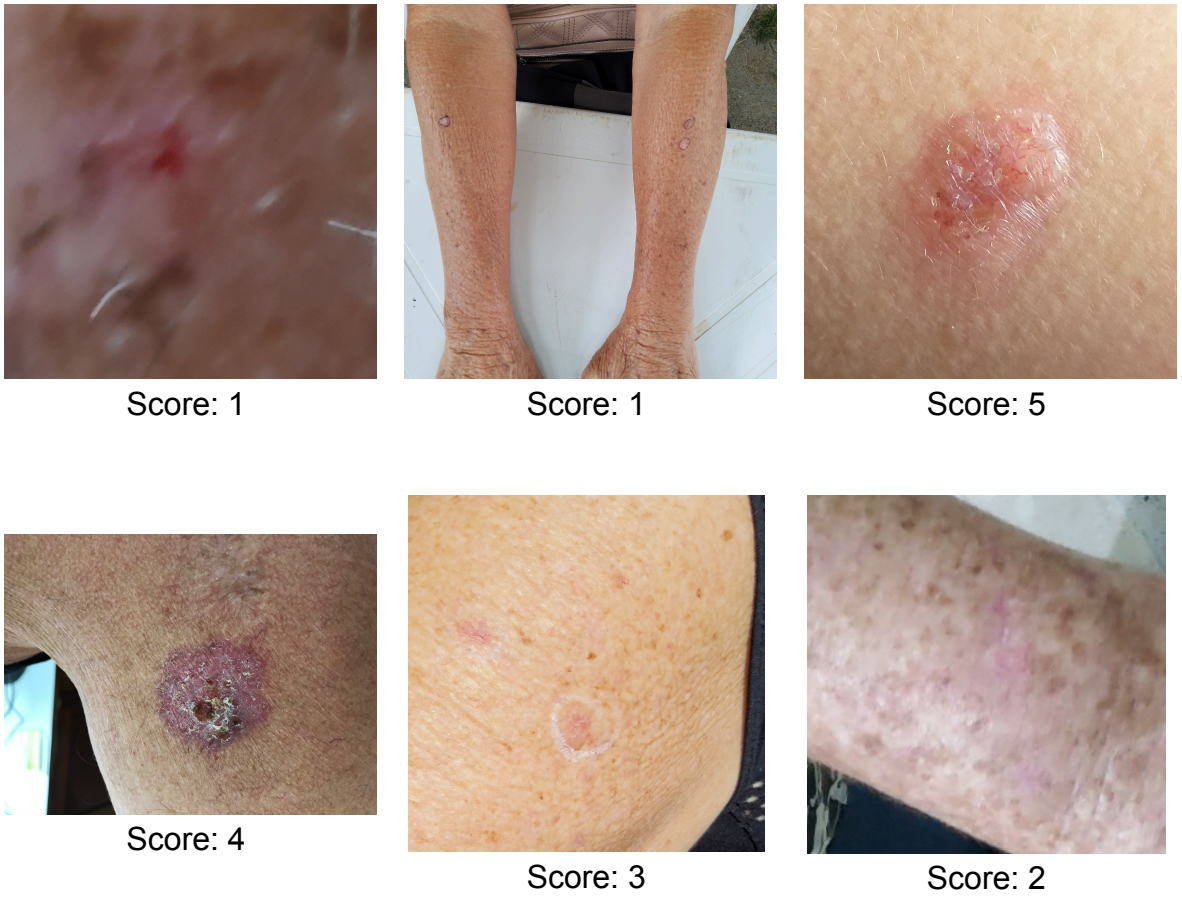
6 - General Perception

This criterion reflects a **general perception** of the image, not focusing on a specific factor. Although subjective, the main consideration is whether the lesion area is clearly and sharply identifiable. If one of the grades is very bad, the general perception will probably be very bad. For example, if the blur grade is 1, it probably influences the image quality a lot, so the general perception grade will probably be 1, even though other criteria receive good grades.

To evaluate this criterion, clinical usefulness (whether a doctor would be able to identify the lesion) can be considered, but the primary focus should be on AI processing - specifically, whether any elements obstruct the lesion and could interfere with automated classification. If the lesion is easily identifiable in the image, the general perception score should be at least 3.

1️⃣ **Very Poor / Inadequate:** unusable for clinical analysis.
2️⃣ **Poor / Below Average / Insufficient:** very limited utility.

3️⃣ **Fair / Acceptable / Average:** usable, but with reservations.
4️⃣ **Good / Adequate / Satisfactory:** reliable for diagnosis.
5️⃣ **Excellent / Great / Ideal:** exemplary image, professional quality.



Example 6: general perception. Examples of scores for this criterion.

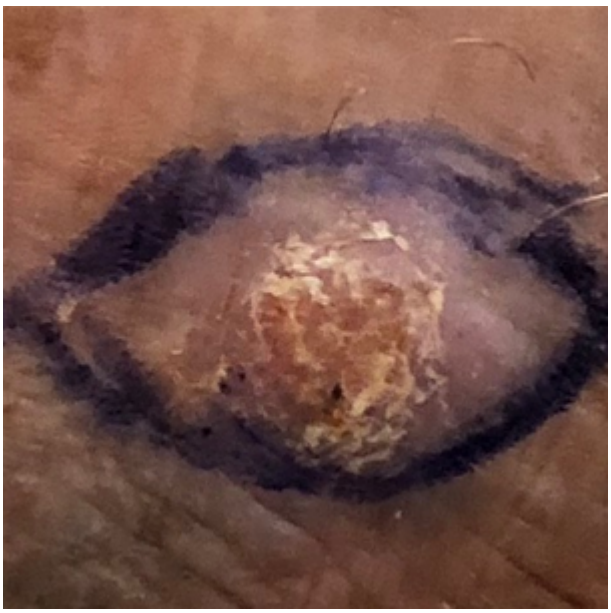
Handling Uncertainties and Ambiguities

Often, criteria may be correlated. For example, distant images are usually poorly framed, meaning these criteria overlap. However, a close-up image can still have poor framing.

In cases where blurriness prevents assessing distance or framing, it is recommended to assign **lower scores** to all affected criteria. In general, individual criteria should be evaluated **independently** whenever possible, and only combined when forming the **general perception** rating.

Annotation Examples

	blur / focus	1
	brightness	2
	distance	5
	framing	2
	interference	5
	general perception	1

	**blur / focus**	2
	brightness	3
	distance	5
	framing	5
	interference	1
	general perception	2

	blur / focus	5
	brightness	5
	distance	1
	framing	1
	interference	1
	general perception	1

	blur / focus	3
	brightness	3
	distance	5
	framing	5
	interference	5
	general perception	3

	blur / focus	5
	brightness	5
	distance	5
	framing	5
	interference	5
	general perception	5

Annotation Documentation

At the end of the process, a document summarizing each image's annotation is expected. Each annotator must complete a document containing the scores for each criterion for each image. Example:

image	blur / focus	brightness	distance	framing	interference	general perception
img1	4	2	5	5	4	3
img2	3	2	4	3	4	3
img3	4	3	5	5	5	4
img4	2	2	5	5	5	3
img5	1	1	5	4	2	1
img6	3	2	5	5	2	3

Example of documentation: illustrative values for demonstration purposes only.